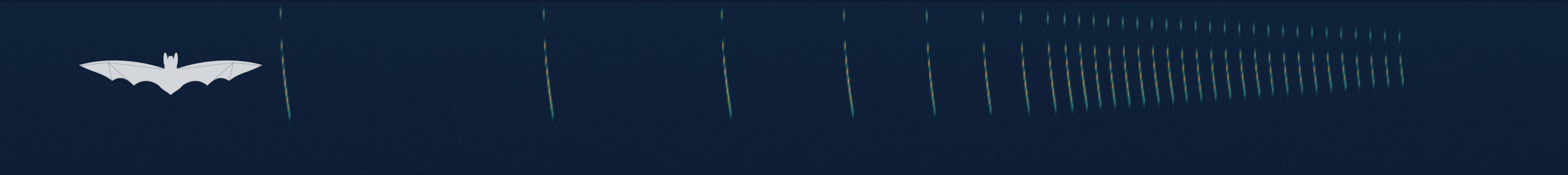
MSc Ecology and Data Science · Dissertation defence

Time-Expanded Perch Embeddings for Feeding-Buzz Detection, Retrieval and Field Candidate Discovery



Candidate WPNS1

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BIOS0057 · University College London · 2025–2026

MOTIVATION

Hours of audio, seconds of signal

9.4 h

of Kenya field audio analysed

270,156

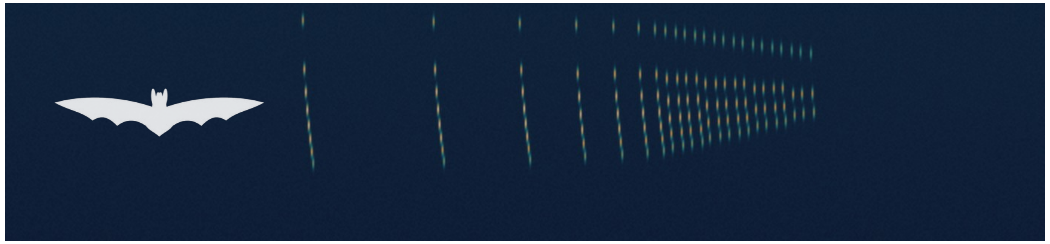
0.25 s windows to be scored

> 200 / s

call rate inside a terminal buzz

The recording scales. The analysis does not.

- Recorders sit in the field for weeks with almost no disturbance to the animals.
- A survey returns many hours of audio for very few events of interest.
- Automated detectors cut that workload — but a detector only ever sees a numerical representation of the sound, and different feature strategies give different answers on the same recordings.



A feeding buzz: search-phase calls compress into a terminal buzz, then stop. Brief, ultrasonic (recorded at 384 kHz), and evidence of foraging — not just of presence.

THE GAP

Detectors exist; transfer is untested

Buzzfindr hand-crafted pulse timing + signal statistics, developed on Ontario recordings. **BatBuddy** a deep-learning object detector over spectrogram images, trained on Dutch recordings.

Both perform well on their own held-out data — and both sets of authors note that broader geographic transfer still needs validation. Overall accuracy can hide large site-to-site differences.

1

RQ1 · Representation

Does a pretrained Perch v2 representation generalise across held-out recording groups better than simpler spectral statistics?

2

RQ2 · Retrieval

How effectively do those representations support held-out-folder retrieval of feeding buzzes from a labelled query?

3

RQ3 · Field transfer

Can a detector built on curated clips support candidate discovery in long, noisy, unlabelled field recordings?

DATA

Curated clips, and continuous field audio

Labelled — Buzzfindr (Ontario, Canada)

All 158 buzz clips + 158 non-buzz clips sampled at random (seed 42) = 316 balanced files. Mono, 16-bit, 384 kHz.

Folder	Buzz	Non-buzz	Total	Location
buzzes_o	21	23	44	Site 3
buzzes_rr	16	25	41	Site 1
buzzes_sp	29	34	63	**Site 4**
buzzes_spmylu	32	26	58	**Site 4**
buzzes_tb	21	22	43	Site 2
buzzes_u	39	28	67	Site 5
Total	**158**	**158**	**316**	**5 sites**

buzzes_sp and buzzes_spmylu are both Site 4 — six folders, five geographic locations.

Field — Mara Triangle, Kenya

- **Site MT18**
  19 Oct – 8 Nov 2019 deployment
- **564 recordings**
  one minute each — 9.4 h analysed
- **1 AudioMoth**
  full-spectrum logger, 384 kHz, gain 1
- **18:00–23:00 & 00:00–05:00**
  EAT coverage — not a full 24-h cycle

No exhaustive ground truth — which bounds every claim I make from this dataset.

Three representations, one detector

Held constant: StandardScaler fitted on training data only → L2-regularised logistic regression (C = 1.0, lbfgs, balanced class weights, threshold 0.5). Any difference in performance is attributable to the representation, not the classifier.

Baseline

1,543

dimensions · spectral statistics

- Temporal mean, SD and maximum for each of 513 frequency bins
- Plus 4 global spectrogram statistics
- Fine frequency detail, fully transparent

Compact

199

dimensions · spectral statistics

- 513 bins collapsed into 64 contiguous frequency bands
- Mean, SD, max per band + 7 global statistics
- Tests whether the extra detail is doing any work

Perch v2

1,536

dimensions · pretrained embedding

- Frozen perch_v2_cpu, no fine-tuning
- Multi-taxa pretraining, built for transfer and similarity search
- Requires an input adaptation — next slide

METHOD · THE KEY ADAPTATION

Making ultrasound legible to a 32 kHz model

SOURCE

384 kHz

Bat calls sit far above 16 kHz.

RESAMPLE

320 kHz

Unpadded waveform resampled, then cropped or padded to 0.25 s = 80,000 samples.

FRONTEND READS

32 kHz

Perch interprets those samples at its own assumed rate.

RESULT

×10

0.25 s is heard as 2.5 s; apparent frequency shifts down tenfold.

Why not simply resample to 32 kHz?

A 32 kHz frontend represents frequencies only up to 16 kHz. Ordinary resampling would discard everything above that — which is the entire signal. Time expansion preserves the relative pulse pattern and moves it into a range the model can process.

Stated as a limitation, not a footnote

Only the Perch route carries this conversion and expansion. What follows therefore compares complete representation pipelines — not embedding architectures under identical preprocessing.

Three levels of difficulty, plus retrieval

1 Stratified random split

222 / 48 / 46 clips. All six folders can appear in each subset — a conventional within-dataset benchmark that does not isolate geography.

2 Grouped-site split

buzzes_o (Site 3) for validation, buzzes_tb (Site 2) for test, 229 clips for training. The test site is geographically separated from training.

3 Leave-one-folder-out ×6

Each folder held out in turn; scaler and detector refitted inside every round. Mean, SD and worst-folder F1 summarise the variation.

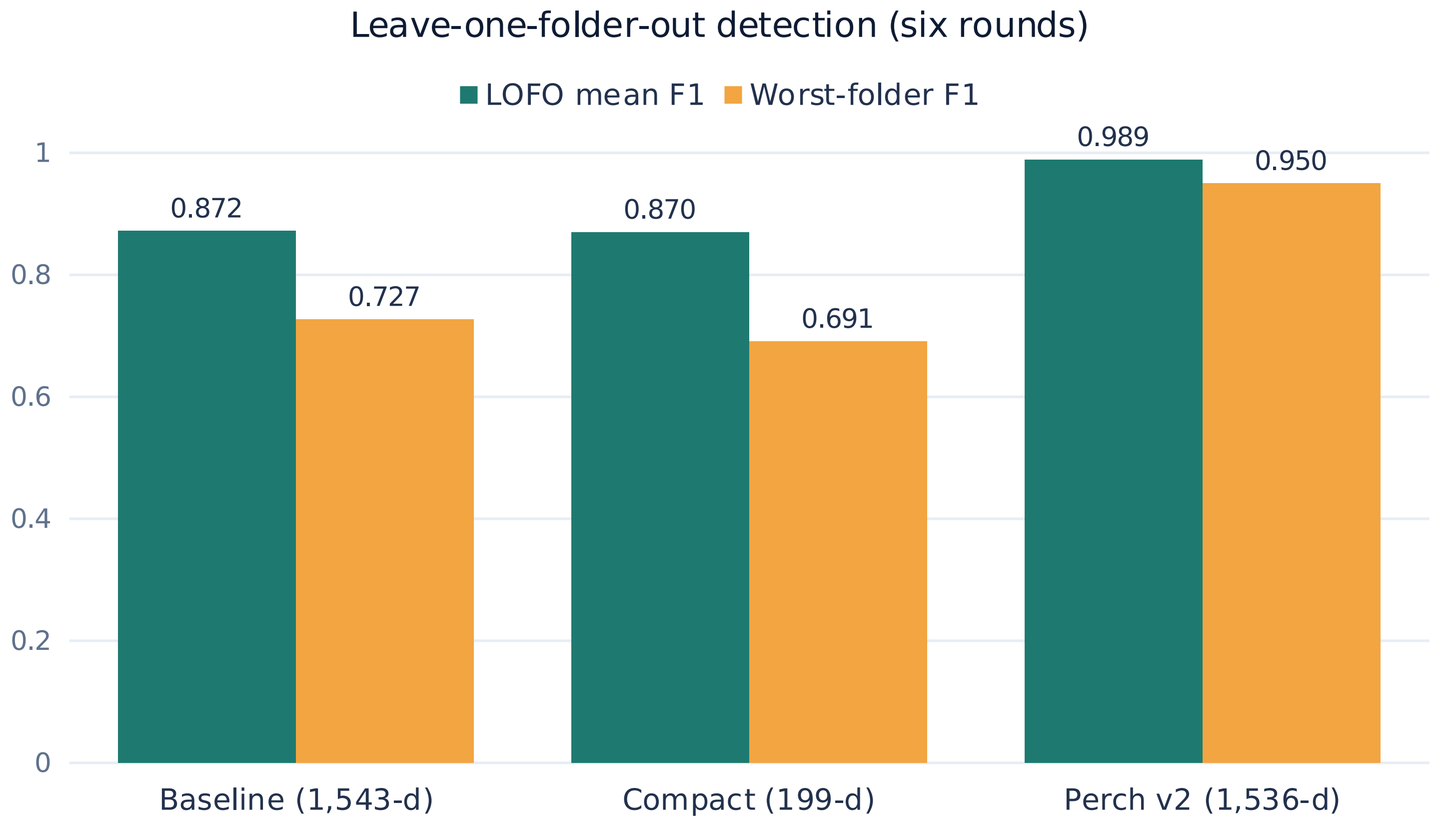
Site-aware retrieval (RQ2)

- All 158 buzz clips used as queries
- The query's own folder excluded from the candidate pool
- Cross-folder pool scaled and L2-normalised, ranked by cosine similarity
- 100 seeded random permutations = an unguided reviewer

Stated up front

buzzes_sp and buzzes_spmylu are both Site 4 — so LOFO and retrieval measure folder-level robustness, not five-location isolation.

Not just better on average — more stable



The grouped-site test was easy

All three representations reached F1 = 1.000 — that split did not discriminate between them.

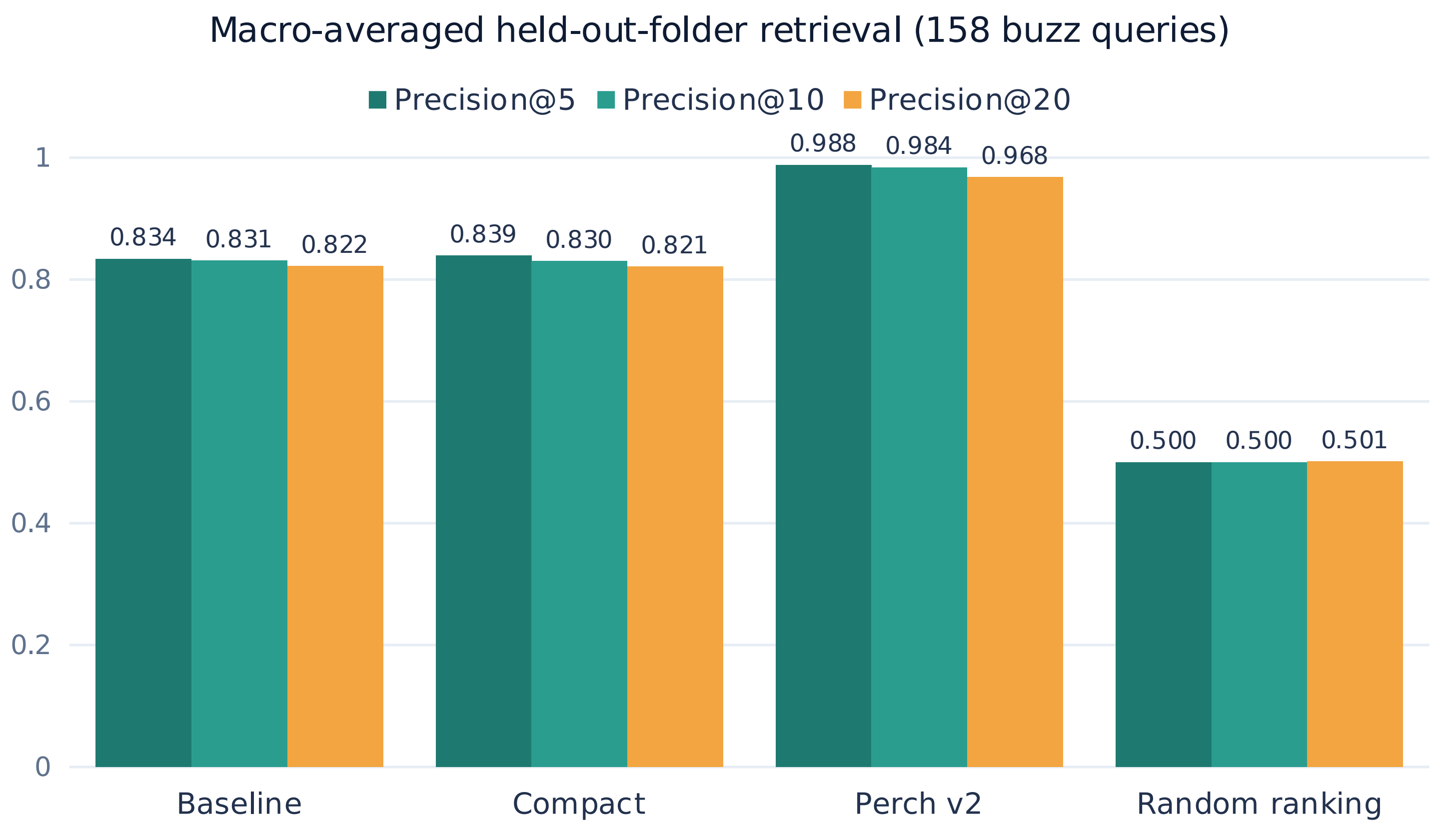
Perch is stable across folders

LOFO SD 0.020, against 0.101 and 0.114. Worst folder 0.950, against 0.727 and 0.691.

More features is not better

199-d compact matched the 1,543-d baseline to within 0.002 F1 — one-eighth of the dimensions.

The same ranking, without a threshold



Average Precision

0.879 Perch v2

Baseline 0.770 · Compact 0.771

Below its own top-k precision — the separation sits near the top of the ranking.

That shape suits a review workflow

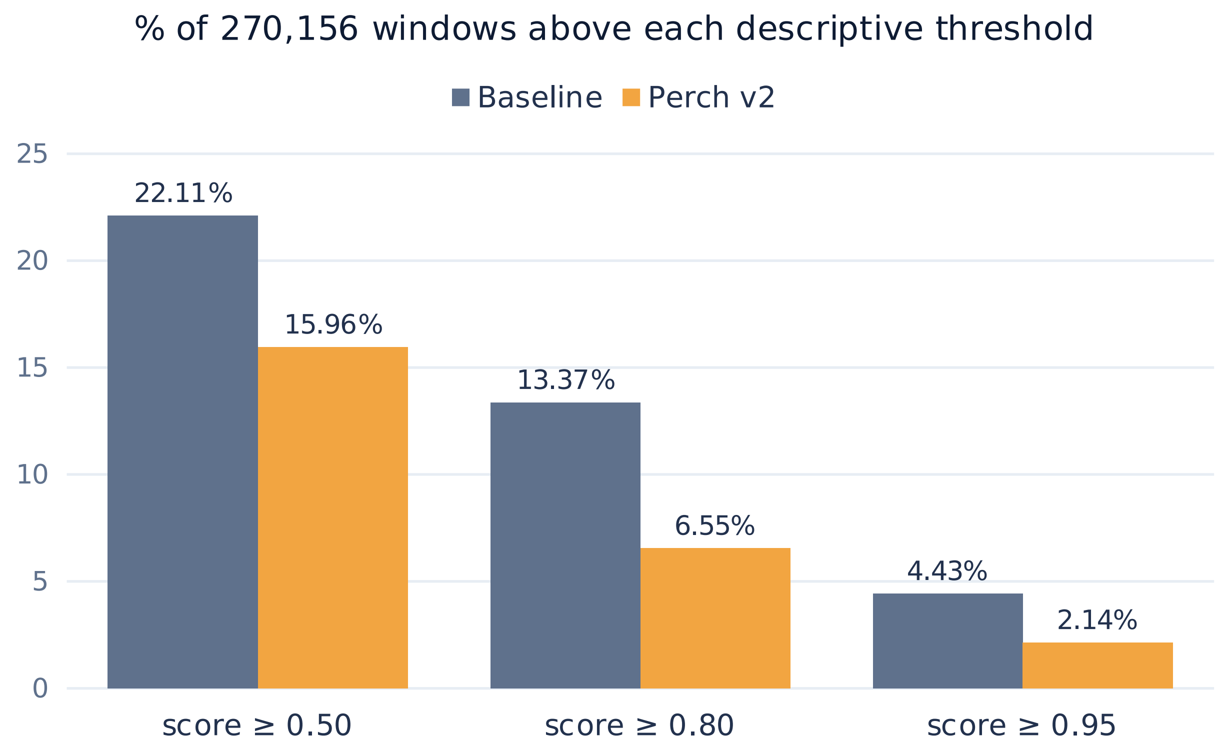
An annotator inspects the highest-ranked candidates first, so precision at the top of the list is what saves time.

An easier problem than the field

The balanced candidate pool puts random Precision@10 at 0.500. Buzzes are far rarer in continuous PAM data.

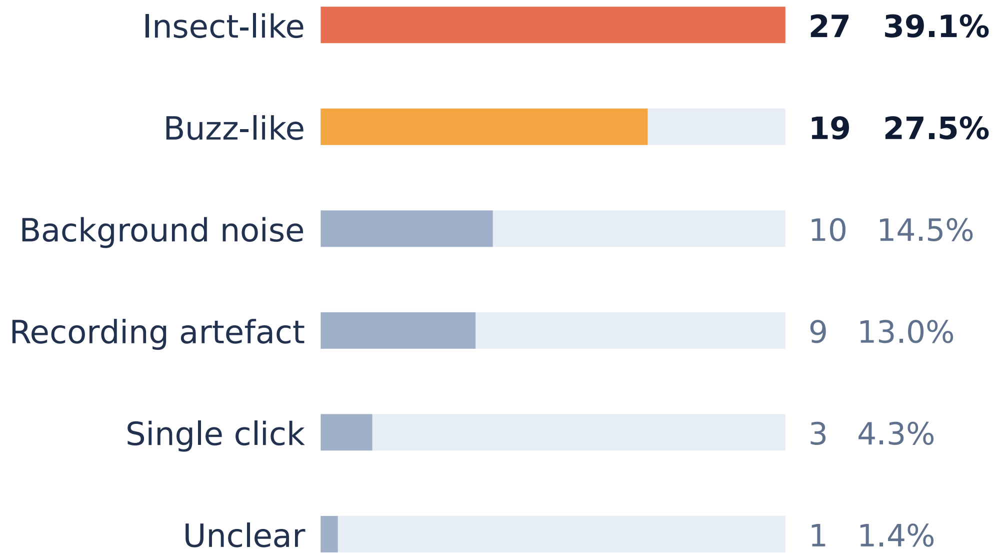
RESULTS · RQ3

Field candidate discovery in Kenya



Different score scales — this compares how each detector prioritised the same recordings, not its precision.

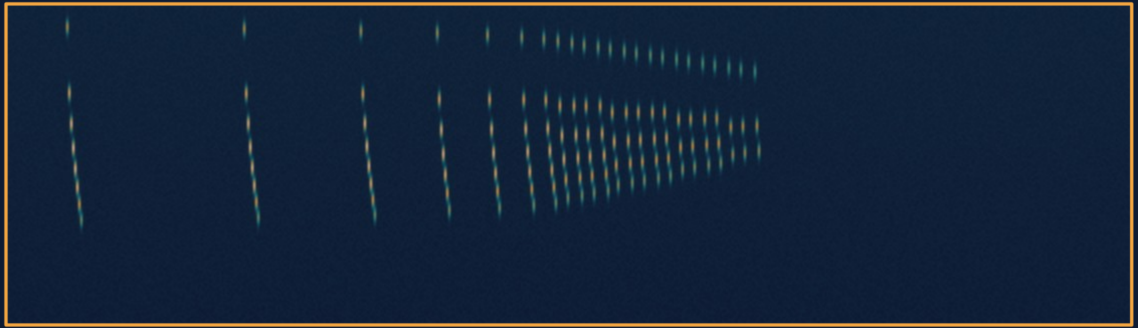
Manual review — 69 Perch candidates, each from a different one-minute recording

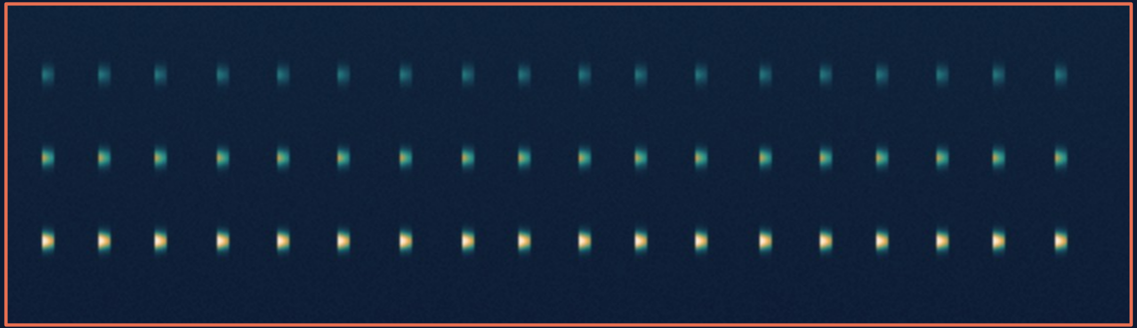


Selected by Perch score — these proportions describe the reviewed candidates, not buzz prevalence across all 270,156 windows.

Inside 0.25 s, these look alike

Schematic illustrations of the two patterns





Buzz-like

Intervals compress into a terminal buzz and the sequence stops. Short, and changing rapidly.

Insect-like

Metronomic intervals that continue for far longer than a buzz ever does. Regular, and persistent.

It is the surrounding second that separates them. Future detectors would benefit from wider contextual windows, or a second-stage model that examines the temporal neighbourhood around a high-scoring short window. The 27 insect-like candidates are ready-made hard negatives for an active-learning cycle.

LIMITATIONS

What I would ask about this work

1 Site 4 folder pairing

buzzes_sp and buzzes_spmylu are the same location, so LOFO and retrieval never fully isolated geography. These values may be optimistic relative to a strict five-location rerun.

The grouped-site test on buzzes_tb does remain geographically isolated.

2 A duration / padding shortcut

At a 0.50 s window, class label aligned perfectly with padded-vs-cropped. I chose 0.25 s to weaken that, but 127 buzz clips were still padded while all non-buzz clips were cropped.

Perch is not immune just because its features are pretrained.

3 Pipelines, not architectures

Only the Perch route carries the 384→320 kHz conversion and tenfold expansion, so the comparison cannot attribute the gain to the embedding alone.

A duration-matched benchmark would separate the two.

4 Kenya has no ground truth

The 69 reviewed candidates were selected by Perch score, so the review proportions are descriptive. No event-level precision or recall can be computed.

Category labels are descriptive, not verified ground truth.

CONCLUSIONS

Representation matters, but is not enough

RQ1

Perch v2 generalised best

After tenfold time expansion, frozen embeddings reached mean LOFO F1 = 0.989 with an SD of 0.020 and a worst folder of 0.950.

RQ2

And retrieved best

Precision@10 = 0.984 against 0.831 and 0.830, with separation concentrated at the top of the ranking, where reviewers actually look.

RQ3

But the field needs more

Perch surfaced buzz-like sequences across many recordings, but insect-like pulse trains were the largest reviewed category. Context and human review still do essential work.

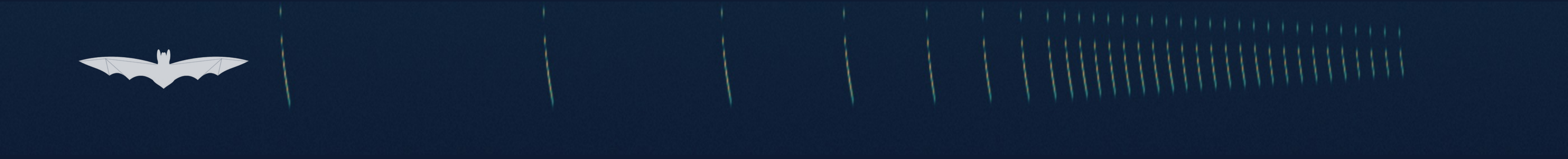
A secondary result: 199 compact dimensions matched 1,543 baseline dimensions to within 0.002 F1 and 0.001 Precision@10. More hand-crafted features are not automatically better.

Next step An independently sampled, exhaustively reviewed Kenya subset — it would give event-level precision and recall, let both detectors be compared on the same confirmed events, and turn those insect-like candidates into hard negatives for retraining.

Thank you

Questions welcome

With thanks to Santiago Martinez Balvanera and Kate Jones for their supervision, and to the researchers who collected and shared the Kenya recordings.



Detection — full metric set

Representation	Random F1	Grouped-site F1	LOFO mean F1	LOFO SD	Worst folder	Mean acc.	ROC-AUC	AP
Baseline (1,543-d)	0.930	1.000	0.872	0.101	0.727	0.883	0.9728	0.9744
Compact (199-d)	0.905	1.000	0.870	0.114	0.691	0.881	0.9716	0.9737
Perch v2 (1,536-d)	**1.000**	**1.000**	**0.989**	**0.020**	**0.950**	**0.990**	**0.9997**	**0.9996**

LOFO mean precision / recall: 0.910 / 0.857 (baseline) · 0.910 / 0.851 (compact) · 0.994 / 0.984 (Perch).

Folder-level range: baseline F1 0.727 (buzzes_sp) → 1.000 (buzzes_tb); compact low 0.691 (buzzes_sp); Perch 0.950 (buzzes_o), 0.983 (buzzes_sp) and 1.000 in the remaining four rounds.

Splits

Random: 222 train / 48 validation / 46 test, stratified. Grouped-site: buzzes_o (Site 3) validation, buzzes_tb (Site 2) test, 229 train. LOFO: six rounds, scaler and detector refitted inside each round. Representation definitions, detector settings and the 0.5 threshold were fixed independently of the reserved validation set.

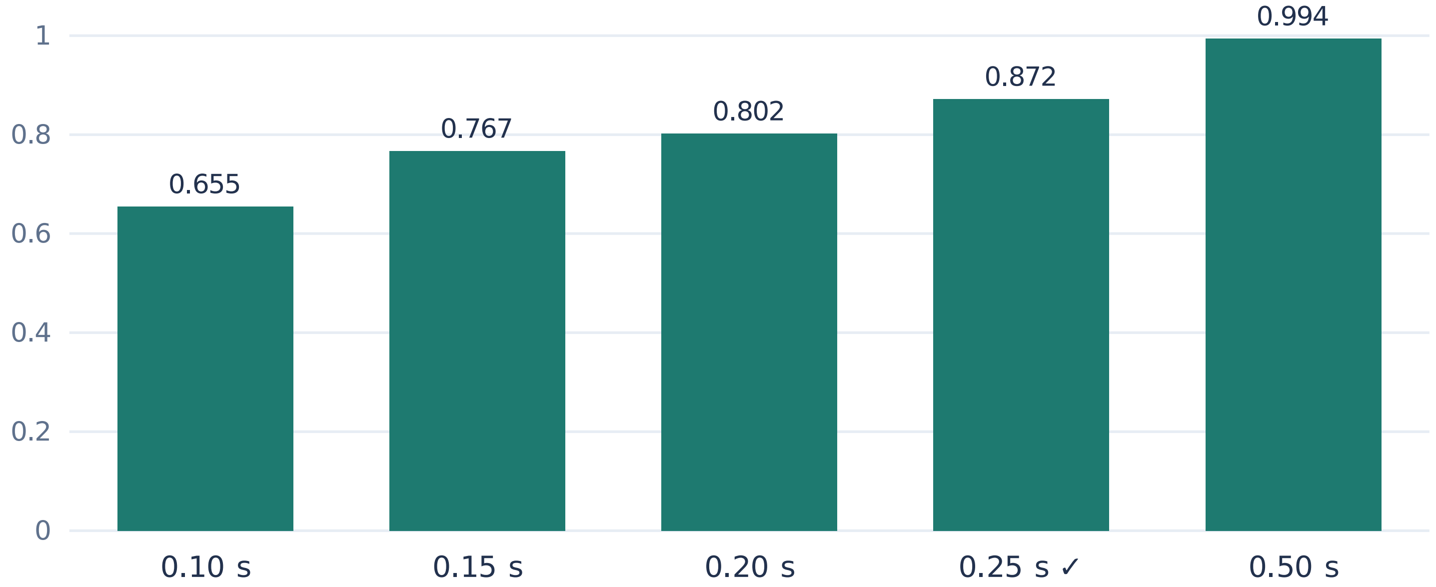
Retrieval — full metric set

Representation	Precision@5	Precision@10	Precision@20	Average Precision
Baseline	0.834	0.831	0.822	0.770
Compact	0.839	0.830	0.821	0.771
Perch v2	**0.988**	**0.984**	**0.968**	**0.879**
Random ranking	0.500	0.500	0.501	0.510

- All 158 labelled buzz clips used as queries; every clip from the query's own folder excluded from the candidate pool.
- StandardScaler fitted on the cross-folder candidate pool only, then applied to both pool and query; vectors L2-normalised and ranked by cosine similarity.
- Folder-level values macro-averaged so that folders with more queries do not dominate.
- Perch Precision@10 ranged 0.948 (buzzes_o queries) to 1.000 (buzzes_tb); folder-level AP ranged 0.750 to 0.955.
- Random baseline: 100 seeded permutations per query (seed 42), generator advancing between repetitions.
- Roughly balanced pools put random Precision@10 at ≈ 0.500 — easier than naturally rare field events.

Preprocessing sensitivity

Analysis-window duration (spectral-statistics detector)



0.50 s scores highest — and is exactly where the padding shortcut is total: all 158 buzz clips padded, all 158 non-buzz clips cropped. 0.25 s was retained as the compromise (127 buzz clips padded, 189 clips cropped). At 0.10 s, random-test F1 was 0.979 but worst-folder F1 collapsed to 0.050.

Spectrogram resolution

- 1024 / 512 (used): LOFO mean 0.872, worst 0.727
- 256 / 128: LOFO mean 0.881, worst 0.714
- 1024 / 900: LOFO mean 0.864, worst 0.691

Settings used throughout

- Periodic Tukey window (α = 0.25), nfft = 1024
- 513 frequency bins × 186 time bins per clip
- log10(magnitude + 1e-10) applied first
- Crop position gave the same pattern

Kenya — thresholds, coverage and timing

Detector	≥ 0.50 windows	≥ 0.80 windows	≥ 0.95 windows	Files with ≥ 0.95
Baseline	59,741 (22.11%)	36,110 (13.37%)	11,978 (4.43%)	161
Perch v2	**43,130 (15.96%)**	**17,692 (6.55%)**	**5,774 (2.14%)**	**121**

Spread across recordings

At the lower two thresholds, Perch windows were spread across more recordings despite being fewer in total — 466 files vs 343 at ≥ 0.50, and 328 vs 254 at ≥ 0.80.

Temporal pattern (EAT, UTC+3)

Highest rates at 23:00 EAT — 30.14% / 17.57% / 7.71% at the 0.50, 0.80 and 0.95 thresholds. Highest daily rates on 24 October 2019. Hours without recordings are not treated as zero.

Candidate selection and review

Route 1 — top 50 Perch windows, neighbouring high-scoring windows within 1 s grouped into one event, one representative window kept per event. Route 2 — eight windows sampled from each of four score bands (0.95–1.00, 0.80–0.95, 0.50–0.80, 0.20–0.50). De-duplicated to one candidate per one-minute recording: 53 retained plus 16 newly reviewed = 69 candidates from 69 distinct recordings. Each reviewed with its 0.25 s spectrogram, 1.25 s of context, pulse-timing structure and time-expanded audio.

Implementation details

Environment

- Python 3.12.13
- SciPy 1.18.0
- scikit-learn 1.9.0
- Perch-Hoplite 1.0.1
- perch_v2_cpu (frozen, no fine-tuning)

Detector

- StandardScaler, fitted on training data only
- Logistic regression, L2 penalty
- C = 1.0, solver lbfgs
- class_weight = balanced
- random_state = 42, max_iter = 1000
- Decision threshold 0.5

Field scoring

- Every 60 s recording peak-normalised
- 0.25 s windows, 0.125 s hop
- 479 windows per recording
- 564 files → 270,156 windows
- Perch: 384→320 kHz, 80,000 samples per window, read as 2.5 s
- Timestamps verified against AudioMoth WAV headers, then converted to EAT